Practice Set: Chain Rule & Automatic Differentiation

Part 1: The Chain Rule (4 Problems)

Problem 1. Differentiate $f(x) = \sqrt{e^{2x} + 4x^3}$ with respect to x. ***Solution:***

We decompose the function into layers: $y = \sqrt{u}$ and $u = e^{2x} + 4x^3$.

Step 1: Outer Derivative

$$y = u^{1/2} \implies \frac{dy}{du} = \frac{1}{2\sqrt{u}}$$

Step 2: Inner Derivative

$$u = e^{2x} + 4x^3 \implies \frac{du}{dx} = 2e^{2x} + 12x^2$$

Step 3: Combine (Chain Rule)

$$f'(x) = \frac{dy}{du} \cdot \frac{du}{dx} = \frac{2e^{2x} + 12x^2}{2\sqrt{e^{2x} + 4x^3}} = \frac{e^{2x} + 6x^2}{\sqrt{e^{2x} + 4x^3}}$$

Problem 2. Differentiate $g(x) = \ln(\sin^2(x) + 1)$. ***Solution:***

Decomposition: $y = \ln(u)$, $u = w^2 + 1$, $w = \sin(x)$.

Step 1: Differentiate components

$$\frac{dy}{du} = \frac{1}{u}$$
$$\frac{du}{dw} = 2w$$
$$\frac{dw}{dx} = \cos(x)$$

Step 2: Multiplication chain

$$g'(x) = \frac{1}{u} \cdot 2w \cdot \cos(x)$$

Step 3: Substitute back

$$g'(x) = \frac{2\sin(x)\cos(x)}{\sin^2(x) + 1} = \frac{\sin(2x)}{\sin^2(x) + 1}$$

Problem 3. Find $\frac{dy}{dx}$ for $y = \cos(x \cdot e^x)$. ***Solution:***

Decomposition: $y = \cos(u)$ and $u = xe^x$.

Step 1: Outer Derivative

$$\frac{dy}{du} = -\sin(u)$$

Step 2: Inner Derivative (Product Rule)

$$\frac{du}{dx} = (x)'e^x + x(e^x)' = 1 \cdot e^x + xe^x = e^x(1+x)$$

Step 3: Combine

$$\frac{dy}{dx} = -e^x(1+x)\sin(xe^x)$$

Problem 4. Differentiate $h(x) = \tan^3(e^{2x})$. ***Solution:***

This is a 3-layer composition: $y = u^3$, $u = \tan(v)$, $v = e^{2x}$.

Step 1: Layer Derivatives

- Outer (u^3): $\frac{dy}{du} = 3u^2$
- Middle ($\tan v$): $\frac{du}{dv} = \sec^2(v)$
- Inner (e^{2x}): $\frac{dv}{dx} = 2e^{2x}$

Step 2: Multiply Chain

$$h'(x) = 3u^2 \cdot \sec^2(v) \cdot 2e^{2x}$$

Step 3: Substitute Back

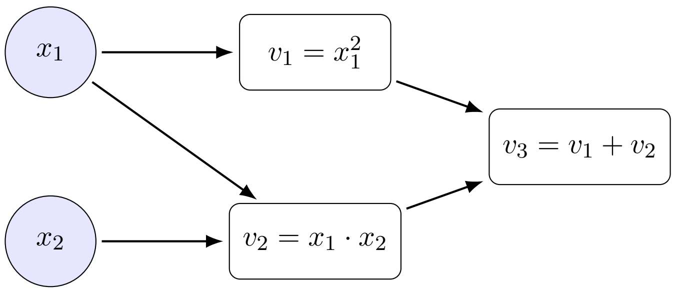
$$h'(x) = 3\tan^2(e^{2x}) \cdot \sec^2(e^{2x}) \cdot 2e^{2x}$$

$$h'(x) = 6e^{2x}\tan^2(e^{2x})\sec^2(e^{2x})$$

Part 2: Automatic Differentiation (4 Problems)

Problem 5. (Forward Mode) Draw the computation graph for $f(x_1, x_2) = x_1^2 + x_1 x_2$. Calculate $\frac{\partial f}{\partial x_1}$ at $(2, 3)$ using the graph. ***Solution:***

1. Computation Graph



2. Primal Trace (Values)

- $x_1 = 2, x_2 = 3$
- $v_1 = 2^2 = 4$
- $v_2 = 2 \cdot 3 = 6$
- $v_3 = 4 + 6 = 10$

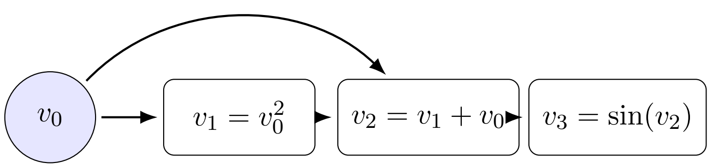
3. Tangent Trace (Derivatives $\dot{v}$) Seed: $\dot{x}_1 = 1, \dot{x}_2 = 0$.

- $\dot{v}_1 = 2x_1\dot{x}_1 = 2(2)(1) = 4$
- $\dot{v}_2 = \dot{x}_1 x_2 + x_1 \dot{x}_2 = (1)(3) + (2)(0) = 3$
- $\dot{v}_3 = \dot{v}_1 + \dot{v}_2 = 4 + 3 = 7$

Answer: Gradient is 7.

Problem 6. (Reverse Mode) Perform Backpropagation on $y = \sin(x^2 + x)$ at $x = 1$ to find $\frac{dy}{dx}$. ***Solution:***

1. Graph Decomposition Nodes: $v_0 = x, \quad v_1 = v_0^2, \quad v_2 = v_1 + v_0, \quad v_3 = \sin(v_2)$.



2. Forward Pass $(x = 1)$ $v_0 = 1 \implies v_1 = 1 \implies v_2 = 2 \implies v_3 = \sin(2)$.

3. Reverse Pass (Adjoints $\bar{v}$) Seed $\bar{v}_3 = 1$.

- $\bar{v}_2 = \bar{v}_3 \cdot \cos(v_2) = \cos(2)$
- $\bar{v}_1 = \bar{v}_2 \cdot 1 = \cos(2)$ (from $v_2 = v_1 + v_0$)

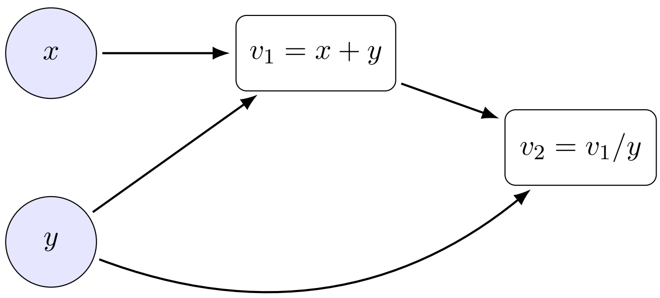
- $\bar{v}_0$ receives gradients from v_1 AND v_2:

$$\bar{v}_0 = (\bar{v}_1 \cdot \frac{\partial v_1}{\partial v_0}) + (\bar{v}_2 \cdot \frac{\partial v_2}{\partial v_0})$$

$$\bar{v}_0 = (\cos(2) \cdot 2v_0) + (\cos(2) \cdot 1) = 3\cos(2)$$

Problem 7. (Reverse Mode - Multi-variable) Calculate ∇f for $f(x, y) = \frac{x+y}{y}$ at $(1, 2)$. ***Solution:***

1. **Graph:** $v_{-1} = x, v_0 = y, v_1 = x + y, v_2 = v_1/y$.



2. **Forward $(1, 2)$:** $v_1 = 3, \quad v_2 = 3/2 = 1.5$.
3. **Reverse (Seed $\bar{v}_2 = 1$):**

- $\bar{v}_1 = \bar{v}_2 \cdot (1/y) = 1 \cdot 0.5 = 0.5$.
- $\bar{y}$ (total) comes from v_1 and v_2:

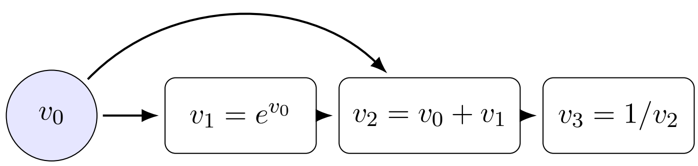
$$\bar{y} = (\bar{v}_1 \cdot 1) + (\bar{v}_2 \cdot \frac{-v_1}{y^2})$$

$$\bar{y} = (0.5) + (1 \cdot \frac{-3}{4}) = 0.5 - 0.75 = -0.25$$

- $\bar{x} = \bar{v}_1 \cdot 1 = 0.5$.

Problem 8. (Reverse Mode) Compute the gradient of $y = \frac{1}{x+e^x}$ at $x = 0$. ***Solution:***

1. **Graph Decomposition** $v_0 = x, \quad v_1 = e^{v_0}, \quad v_2 = v_0 + v_1, \quad v_3 = 1/v_2$.



2. **Forward Pass** $(x = 0)$

- $v_0 = 0$
- $v_1 = e^0 = 1$
- $v_2 = 0 + 1 = 1$

- $v_3 = 1/1 = 1$

3. Reverse Pass (Seed $\bar{v}_3 = 1$)

- $\bar{v}_2 = \bar{v}_3 \cdot (-1/v_2^2) = 1 \cdot (-1) = -1$
- $\bar{v}_1 = \bar{v}_2 \cdot 1 = -1$
- $\bar{v}_0$ (total) $= (\bar{v}_2 \cdot 1) + (\bar{v}_1 \cdot e^{v_0})$

$$\bar{v}_0 = (-1) + (-1 \cdot 1) = -2$$

Answer: $\frac{dy}{dx} = -2$.